



Soulprint

A living AI dossier for your wallet.

Mint once. It evolves forever.

Every wallet has personality. Nobody can read it.

On-chain history holds an identity in raw counters — transactions, deploys, NFT mints, gas spent. But it's locked in numbers no human reads, and no contract composes on. Meanwhile most 'AI on-chain' projects today are smoke and mirrors: they call Gemini or Groq off-chain and only settle the result on-chain. The AI isn't really on-chain.

O

on-chain primitives
that compose on a wallet's
identity today



An on-chain identity primitive, written by on-chain AI agents.



- 1 _____ Point Soulprint at any wallet — yours, or someone else's.
- 2 _____ A Somnia JSON API agent reads its real on-chain activity.
- 3 _____ A Somnia on-chain LLM (Qwen-3 30B, fixed seed, validator-consensus) writes a witty, structured dossier.
- 4 _____ It mints as a soulbound NFT (ERC-5192). Can't be sold. Can't be transferred.
- 5 _____ An on-chain Cron keeps re-evaluating it — autonomously, forever.

End to end — about six seconds. AI executed by validators, output verified by consensus.

Try it. Now.

Everything below is live on Somnia Shannon testnet (chainId 50312).



FRONTEND

<https://soulprint-psi.vercel.app>

Mint a soulprint, look up any wallet.

GITHUB

<https://github.com/DontFoldBB/soulprint>

Public repo. 47 Hardhat tests green.

SOULPRINT

0x6876041cc67f9cd1b11e6e1827b13f3622d256e5

ERC-721 + ERC-5192 soulbound, dynamic on-chain tokenURI.

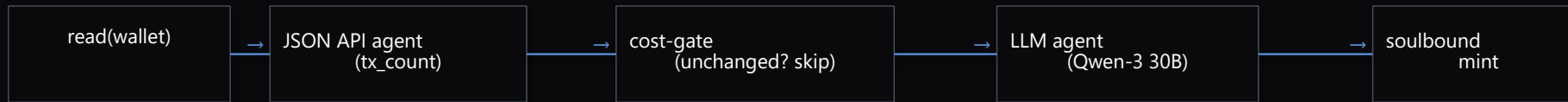
SOULPRINTCRON

0x0bf4e395ad3746632f86b5254fa18f0db3479d95

Somnia Reactivity handler. Self-rescheduling autonomy.

Architecture

A two-step on-chain agent pipeline with a cost-gate in the middle, an autonomy loop above.



Autonomy

SoulprintCron → tick → evolveBatch() → self-reschedule (no human tx between ticks)



The Soul Evolution System

On-chain visual evolution — derived from real activity, not from the LLM's mood.



10 stages × 3 archetype lines = 30 spirit forms



- Stage 1-10 bucketed from on-chain tx_count. Stage 4 at 75 tx, Stage 10 at 200k.
- Form lookup = (archetype × stage) → one of 30 monochrome spirit PNGs.
- Stage and Form are stored on-chain — stageOf / formIdOf / formSlugOf / evolutionOf — and surfaced as tokenURI traits.

Prepaid Evolution Fuel

Solves the 'autonomous tick eats the project's reserve forever' problem, on-chain.

BEFORE

One mint → autonomous evolution forever, from the project's pocket.

100 active wallets $\approx$ 100 STT / day burn.

Owner can withdraw the entire reserve — no protection for funded evolutions.

NOW (LIVE)

- Mint earmarks INITIAL_FUEL_GRANT (0.4 STT $\approx$ 1 evolution) into the token's evoBalance.
- Each successful re-evolution deducts EVOLUTION_COST. Out of fuel → soul freezes (EvolutionPaused).
- ANYONE can call topUpEvolution(tokenId). Owner cannot withdraw it (gated by availableForWithdraw; totalReserved is sacred).

*Keeping a soul alive becomes a public good.
Boost a friend. Boost a stranger.*



Autonomy — proven live on-chain

2 real evolutions

33 cost-gated skips

0 human tx

35

autonomous ticks on the previous live cron

SoulprintCron is a Somnia Reactivity SomniaEventHandler. On each scheduled tick the handler calls Soulprint.evolveBatch() and then schedules the next tick — inside the same handler. The chain ticks the contract for us, on time, forever. Sender of every tick is the Reactivity precompile (address 0x...0100). The contract owner's address has zero outgoing tx to the cron — fully verifiable on the explorer. Fresh cron at 0x0bf4...9d95 is armed now; same code as the proven one — criterion holds.



It's not an NFT. It's a primitive other agents read.



ONE-CALL READS —

profileOf, traitsOf, evolutionOf, evolutionFuel. Any contract or agent gets a wallet's Soulprint in one read.

MCP SERVER —

node mcp/dist/index.js exposes get_soulprint(wallet) over the Model Context Protocol. Any AI agent can read a wallet's identity.

EXAMPLEGATE —

a separate contract whose enter() succeeds only if the caller's Soulprint activityScore clears a threshold. Reputation that can't be bought, transferred, or faked.

How we map to the 4 judging criteria

- | | | |
|----|--|---|
| 1. | FUNCTIONALITY | Live on Somnia testnet. read → AI → soulbound mint pipeline runs end-to-end in ~6 s. 47 Hardhat tests green. |
| 2. | AGENT-FIRST DESIGN | Orchestrates TWO Somnia base agents (JSON API + LLM) per pipeline. profileOf / traitsOf / evolutionOf / evolutionFuel for any contract to read. MCP server + ExampleGate as live consumers. |
| 3. | INNOVATION & TECHNICAL CREATIVITY | Native on-chain inference (not off-chain).
Chained read→reason agent pipeline. Soulbound IDENTITY (not collectible).
On-chain dynamic tokenURI. Cost-gated evolution. 30-form Soul Evolution System.
Prepaid Evolution Fuel. |
| 4. | AUTONOMOUS PERFORMANCE | SoulprintCron self-reschedules INSIDE the handler.
ticks() rises, subscriptionId re-arms, zero human tx between ticks. Previous cron:
35 ticks → 2 real evolutions, all verifiable on explorer. |

Tech stack & numbers

47

Hardhat tests

30

spirit PNG forms

10

evolution stages

~6s

end-to-end mint

0

human tx per evolution

Solidity 0.8.30 · OpenZeppelin v5 · ERC-721 + ERC-5192 · viaIR · cancun EVM ·
Hardhat + viem · Somnia Agents · Somnia Reactivity · Qwen-3 30B (on-chain) ·
Model Context Protocol · Next.js 16 · Turbopack · Vercel · TypeScript 5.7

Soulprint

A composable identity primitive other agents read.

Cost-bounded by design. Soulbound. Self-evolving. Live on Somnia.

Mint once. It evolves forever.

